

120 Days of Quant Finance

Week 7: Time Series Analysis Basics

Day 43: Introduction to Time Series

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1. What is a Time Series?

A time series is a sequence of observations indexed by time:

$$\{X_t\}_{t=1}^T$$

Unlike standard random variables, the order of observations matters. Time series data often exhibit dependence between current and past values.

2. Time Series vs Independent Observations

In classical probability theory, observations are typically assumed to be independent.

In time series analysis:

$$X_t \text{ may depend on } X_{t-1}, X_{t-2}, \dots$$

This temporal dependence is the defining feature of time series data.

3. Financial Time Series

In finance, we often model returns instead of prices.

If S_t is the asset price at time t , the log-return is:

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right)$$

Returns are generally more stable and suitable for statistical modeling than prices.

4. Components of a Time Series

A time series may contain:

- **Trend:** Long-term movement in the data.
- **Seasonality:** Regular repeating patterns.
- **Noise:** Random fluctuations.

5. Deterministic and Stochastic Trends

Deterministic Trend

A deterministic trend can be modeled as:

$$X_t = a + bt$$

Stochastic Trend (Random Walk)

A stochastic trend follows:

$$X_t = X_{t-1} + \epsilon_t$$

where ϵ_t is white noise.

This structure is common in financial markets.

6. Importance in Quantitative Finance

Time series analysis allows us to study:

- Predictability of returns
- Volatility persistence
- Mean reversion
- Random walk behavior

It forms the foundation for volatility modeling, forecasting, and quantitative trading strategies.

Conclusion

Time series analysis studies how randomness evolves over time. Understanding temporal dependence is essential for modeling financial markets and risk dynamics.